

Jorge Manuel Torre

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Education

Virginia Tech

Blacksburg, VA

B.S. Computer Science, Overall GPA: 3.32/4.0, Major GPA: 3.50/4.0

May 2026

Relevant Coursework: Data Structures & Algorithms, Computer Systems, Computer Architecture, Software Reverse Engineering, Linux Kernel Programming, Security of AI.

Projects & Experience

Garage64: Custom Entity Model Studio • GitHub

August 2025 – Present

React, Three.js, Django, SQLite, Docker

- Built a self-hosted web app for creating custom Minecraft car models (OptiFine CEM) and texture packs, packaged with Docker Compose for local setup.
- Created a React and Three.js 3D workspace where users can swap vehicle parts and apply custom paint textures in real time.
- Built a Django REST API for entity data and a one-click export pipeline for .jem and .jpm asset files.

personal-blog: Full-Stack Blogging Platform • Website

February 2026 – April 2026

React, Vite, Django REST Framework, PostgreSQL, Docker, Nginx

- Built a blogging platform with a Django REST API, React + Vite frontend, Docker Compose setup, and Nginx reverse proxy.
- Added JWT-authenticated admin access, draft and publish workflows, cover image uploads, and live markdown previews with syntax highlighting.
- Exposed a public read-only API so posts can appear on a separate portfolio site while the main blog runs at blog.sys32ent.com.

CHUD: Catastrophic Harmful Update Detection • GitHub

January 2026 – April 2026

Python, PyTorch, LoRA, Llama-2-7b, Llama-Guard-3-8b, Unsloth

Team of 4 (CS Capstone)

- Tested the LoX defense (Perin et al., COLM 2025) against an iterative LoRA fine-tuning attack on Llama-2-7b using benign GSM8k and adversarial BeaverTails samples.
- Built an Attack Success Rate (ASR) evaluation pipeline with Llama-Guard-3-8b as the judge model and ran 30+ experiments across attack mixes and learning rates.
- Found that LoX lowered ASR from 91–96% to 37–63% under direct harmful fine-tuning, but an iterative mixed-data attack raised ASR back to 89%.
- Set up x86-64 and aarch64 VT GPU cluster environments for the team and ran QLoRA fine-tuning evaluations for the final analysis.

Celeste-RL: Deep Reinforcement Learning for a Precision Platformer • GitHub • Website

January 2026 – April 2026

Python, PyTorch, NumPy, matplotlib, CUDA

- Compared five RL methods (DQN, Behavioral Cloning, Hybrid, Curriculum, Dueling DQN with curiosity) trained to play Celeste Classic on the Pyleste emulator.
- Built scripts for batched evaluation, comparison plots, gameplay GIF generation, and a GitHub Pages results site.

Experimental Home Lab

January 2021 – Present

Proxmox, Debian, Ubuntu, NixOS, Docker

- Run a Proxmox home lab with an OPNsense firewall, SSH jumpbox access, and NixOS / Docker environments for self-hosted services and systems experiments.

Leadership & Community

Team Lead, Database Management Systems (CS 4604)

May 2025 – August 2025

- Led a 3-person team through schema design and SQL development, then integrated the team's work into the final submission.

Team Lead, Intermediate Software Design (CS 3704)

January 2025 – April 2025

- Led a 5-person team through requirements gathering, sprint planning, code reviews, and final project delivery.

Garage32 (Community Project)

March 2020 – December 2021

- Led development and distribution of a graphical asset pack for the Minecraft community, reaching **600+ downloads**; promotional content earned **30K+ likes** on a single TikTok post.

Skills

Languages: Python, JavaScript, TypeScript, Java, C, C++, SQL, Bash

Web & Frameworks: React, Three.js, Vite, Django, Django REST Framework, Node.js, REST APIs

Infrastructure: PostgreSQL, Docker, Nginx, Linux, Git, CI/CD, NixOS, Proxmox